IBM University Engagement Project Submission

“Exploring the Power of Machine Learning with IBM Cloud”

Project Tile – AI-Powered Predictive Maintenance of Industrial Machinery using IBM watsonx.ai

Domain of Project – Manufacturing / Industrial AI

Student Name	Recent Passport Photo	Email ID	Mobile number [WhatsApp]
Abhay Rajendra Gaikwad		**abhaygaikwad112005@gmail.com / 202401100146@mitaoe.ac.in**	**8446075447**

Problem Statement -

Problem Statement: Predictive Maintenance of Industrial Machinery

Modern industries rely heavily on continuous machine operation, making unexpected equipment failures a major cause of production delays, increased maintenance costs, and reduced operational efficiency. Traditional maintenance strategies, such as reactive repairs or scheduled servicing, often fail to detect potential faults before breakdowns occur. This project aims to develop an AI-powered predictive maintenance system using IBM watsonx.ai AutoAI that analyzes machine sensor data, including air temperature, process temperature, rotational speed, torque, tool wear, and machine type, to accurately predict different types of machine failures. By identifying failures such as Tool Wear Failure, Heat Dissipation Failure, Power Failure, Overstrain Failure, Random Failure, or No Failure in advance, the system enables proactive maintenance planning, minimizes unplanned downtime, improves equipment reliability, optimizes maintenance costs, and supports smarter decision-making in Industry 4.0 manufacturing environments.

Proposed Solution -

(IBM watsonx.ai Studio – AutoAI):

We developed an AI-powered Predictive Maintenance System using IBM watsonx.ai Studio and AutoAI.

- Machine sensor data is uploaded into IBM watsonx.ai.
- AutoAI automatically performs data preprocessing and feature engineering.
- Multiple machine learning pipelines are generated and evaluated.
- Hyperparameter optimization is applied to improve model performance.
- The best-performing Random Forest model is selected automatically.
- The deployed model predicts machine failure types using real-time sensor values.
- The solution enables proactive maintenance, reduces downtime, and improves equipment reliability.

Technology Used

IBM Cloud Services :

- IBM Cloud Lite
- IBM watsonx.ai Studio
- IBM AutoAI
- IBM Watson Machine Learning
- IBM Cloud Deployment Space

Programming Language : Python

Dataset : Kaggle Predictive Maintenance Dataset

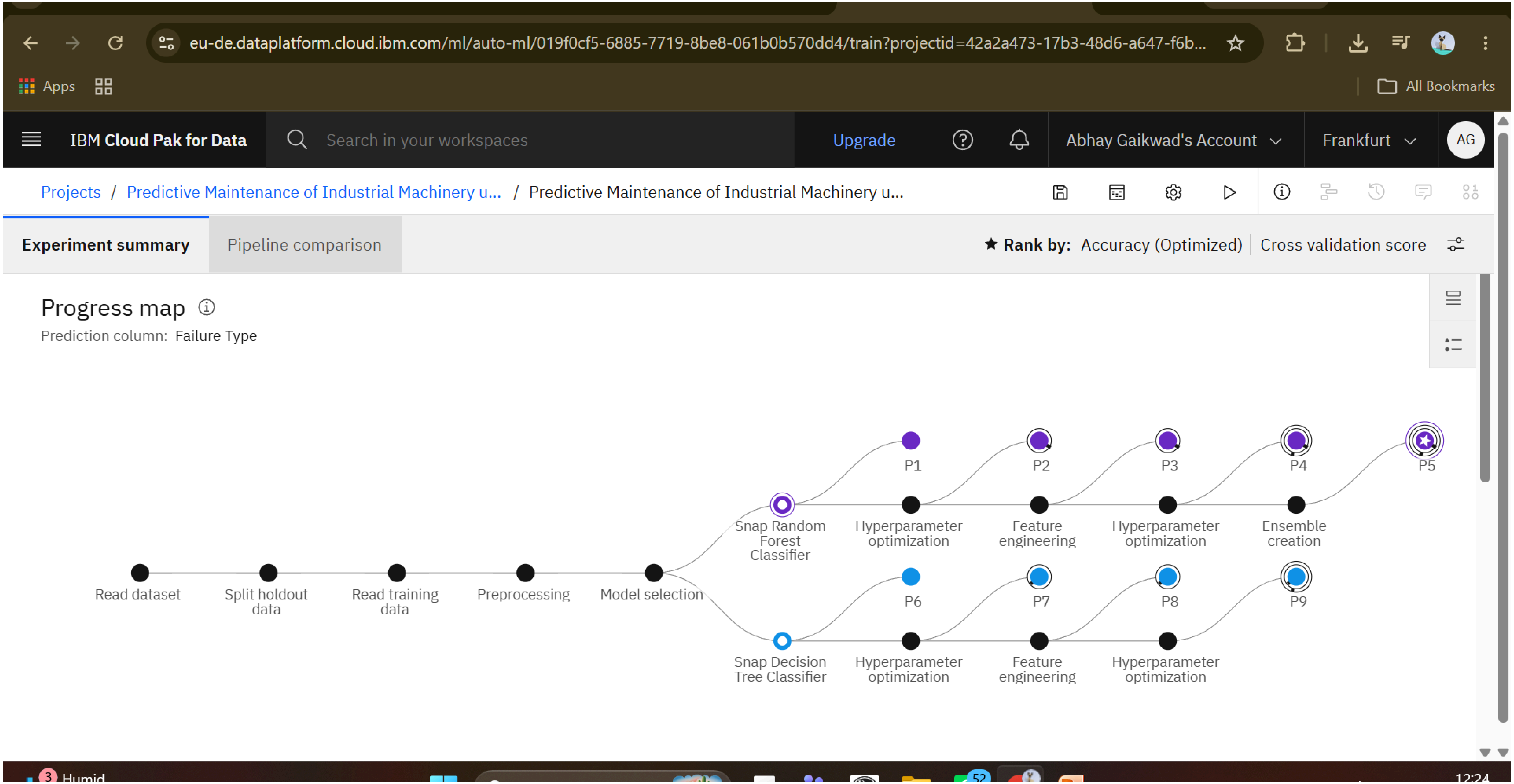
Machine Learning Algorithm : Random Forest Classifier (AutoAI Selected)

Accuracy Obtained : 99.5%

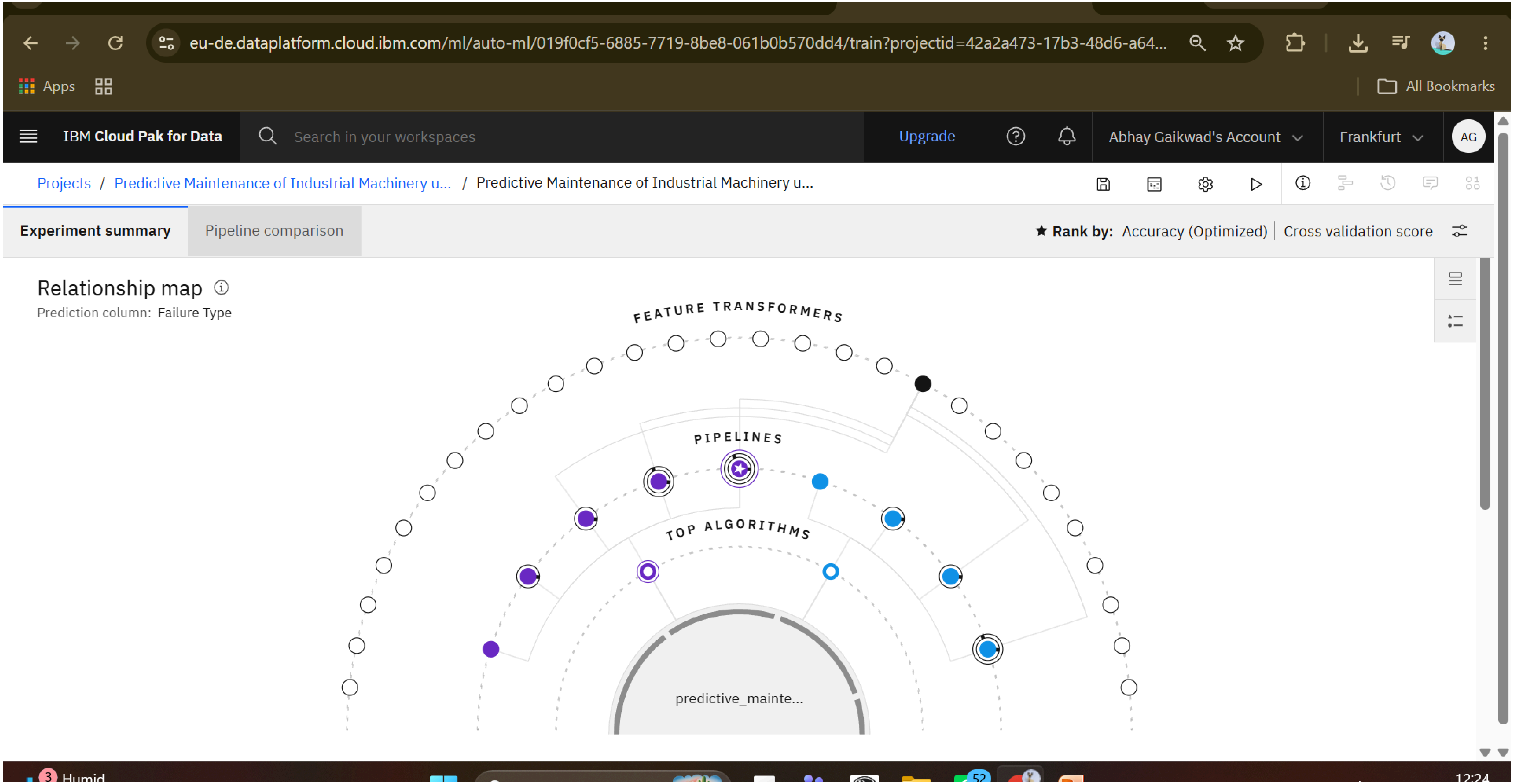
Enhancements :

- Automatic Data Preprocessing
- Feature Engineering
- Hyperparameter Optimization (HPO)
- Batch Optimization
- Ensemble Learning

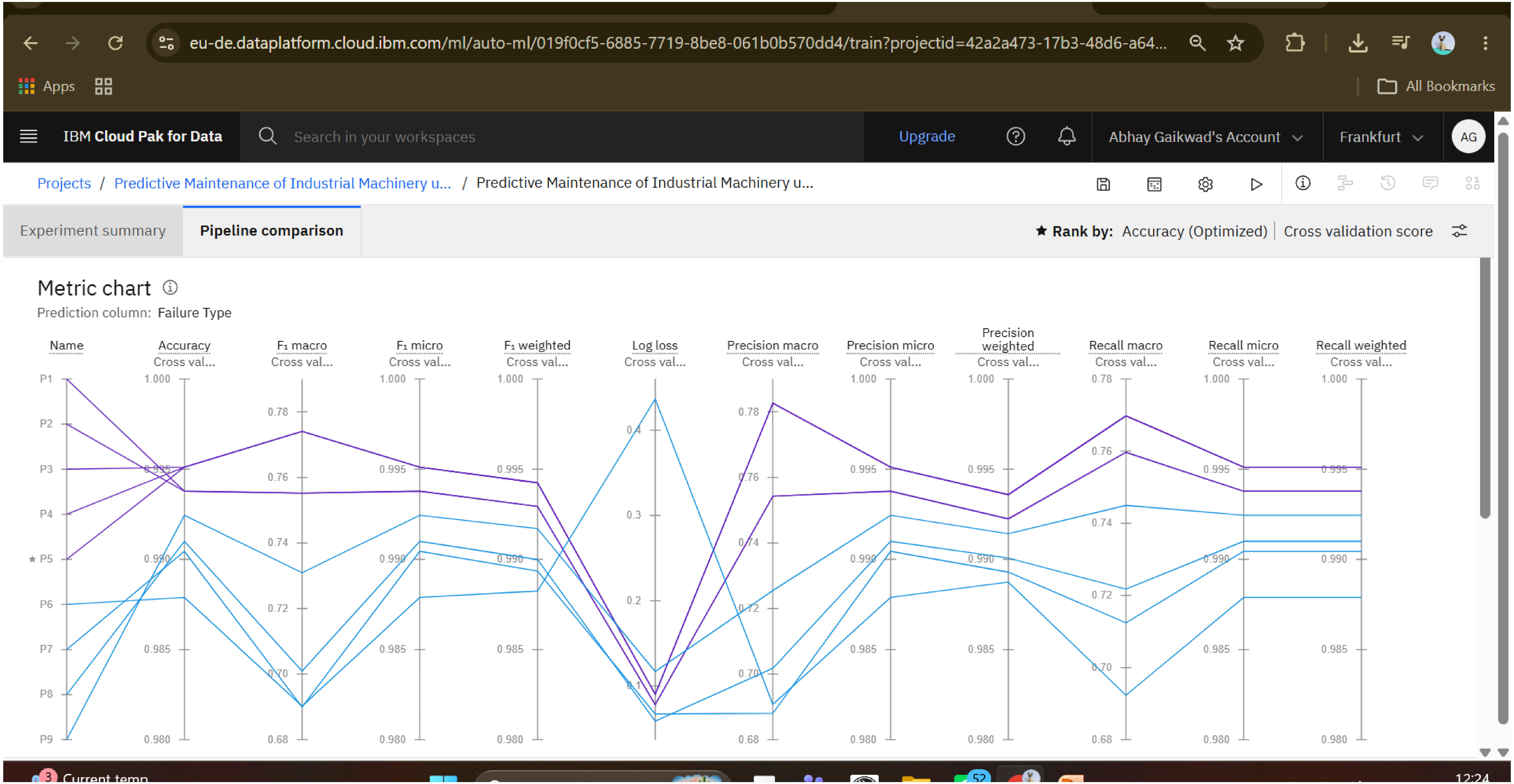
Training Screenshots-



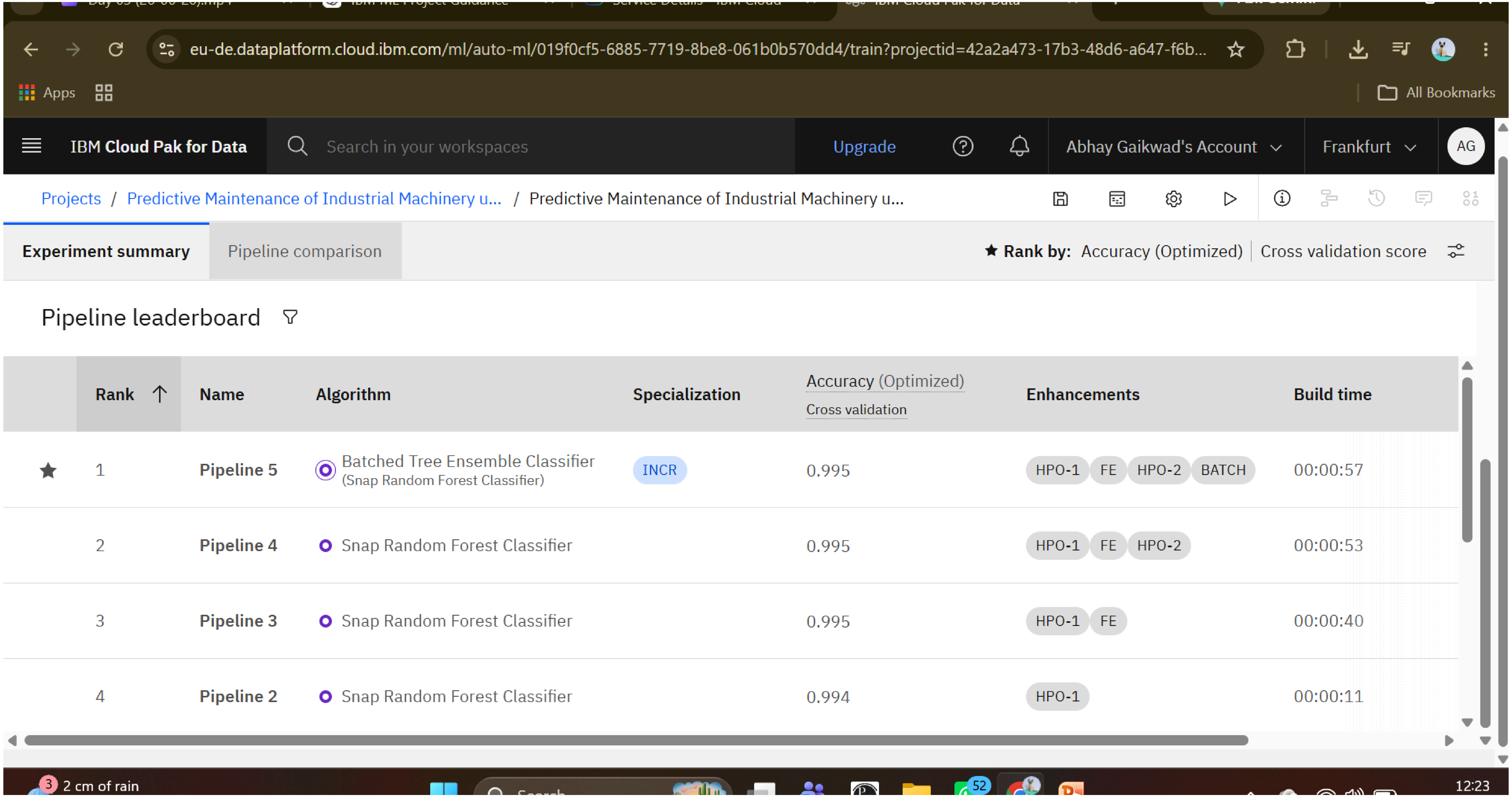
Training Screenshots-



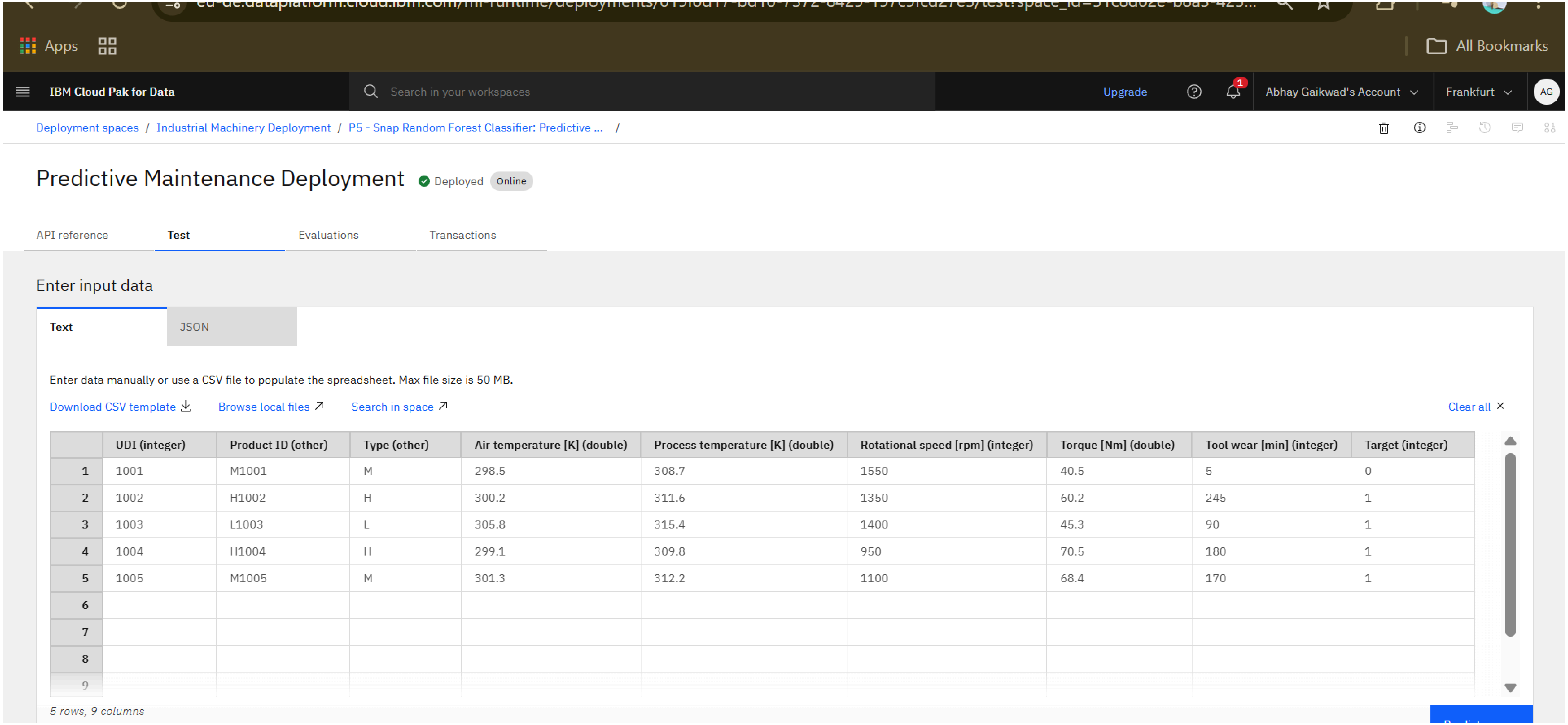
Training Screenshots-



Best Model Screenshot:

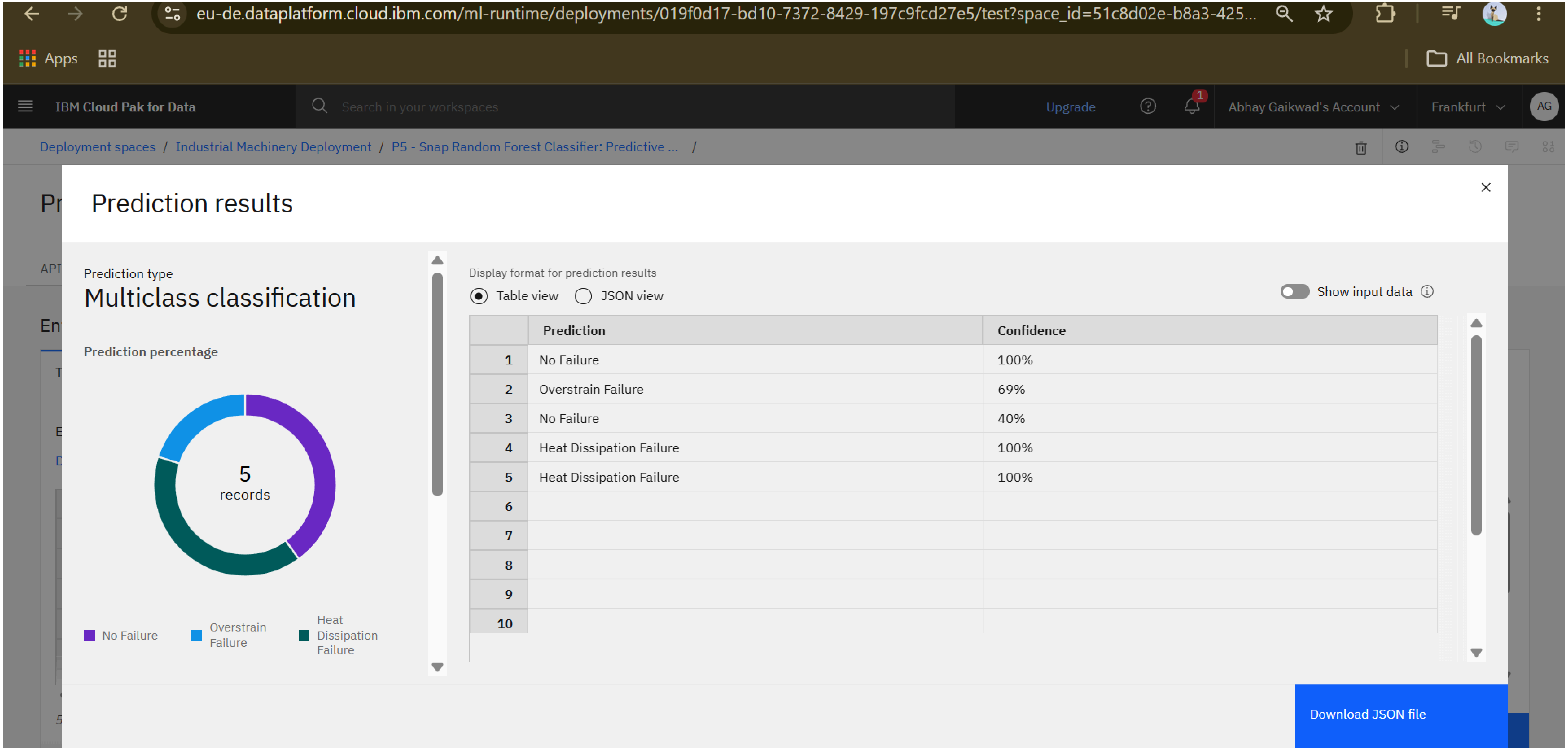


Testing Screenshot:



	UDI (integer)	Product ID (other)	Type (other)	Air temperature [K] (double)	Process temperature [K] (double)	Rotational speed [rpm] (integer)	Torque [Nm] (double)	Tool wear [min] (integer)	Target (integer)
1	1001	M1001	M	298.5	308.7	1550	40.5	5	0
2	1002	H1002	H	300.2	311.6	1350	60.2	245	1
3	1003	L1003	L	305.8	315.4	1400	45.3	90	1
4	1004	H1004	H	299.1	309.8	950	70.5	180	1
5	1005	M1005	M	301.3	312.2	1100	68.4	170	1
6									
7									
8									
9									

Testing Screenshot:



Novelty and Uniqueness

- Uses IBM watsonx.ai AutoAI to automate machine learning model development.
- Automatically identifies the most suitable classification algorithm.
- Predicts failures before machine breakdown occurs.
- Supports predictive maintenance instead of reactive maintenance.
- Reduces maintenance costs and unexpected downtime.
- Improves industrial productivity and operational efficiency.
- Easily scalable for Industry 4.0 and Industrial IoT environments.
- Helps organizations optimize resource utilization, minimize operational risks, and improve production planning.
- Demonstrates the practical application of Artificial Intelligence and Machine Learning in smart manufacturing environments.

Git Hub Link

Please create public github repository (Enable Add README button while creating repository) with format –

Please test and Paste the working Github URL –

https://github.com/AbhayGaikwad05/predictive-maintenance-ibm-watsonx.git

Make sure you have Uploaded following three files into your github repository

1) Data.csv file, 2) yourproblemstatement.pdf file 3) your project presntation.pptx file 4) Python Notebook downloaded from IBM Cloud

Future Scope

- Enable live streaming prediction using IBM Cloud.
- Develop a predictive maintenance dashboard.
- Extend the model for Remaining Useful Life (RUL) prediction.
- Integrate with factory ERP systems.
- Deploy on edge devices for smart manufacturing.
- Improve prediction using deep learning techniques.
- Improve prediction using deep learning techniques.
- Enhance cybersecurity by integrating secure IoT communication and access control for industrial environments.
- Combine machine vision with sensor data to improve fault detection through multimodal AI.
- Support sustainability initiatives by reducing energy consumption, minimizing equipment waste, and extending machine lifespan through optimized maintenance.

Certificates Earned:

Please add your credly certificate

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IBM SkillsBuild

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This certificate is presented to

Abhay Gaikwad

for the completion of

Lab: Troubleshoot Your Code Using IBM Bob

(ALM-COURSE_4071307)

According to the Adobe Learning Manager system of record

Completion date: 19 Jun 2026 (GMT) **Learning hours:** 30 mins

Thank You!

Thank you for your time and interest.